PARTIAL- SPECIAL MATHEMATICS

Linear and Homogeneous SDEs

Algorithm:

1. Find the eigenvalues of A using the formula $det(A - \lambda I_n) = 0$

   Find $\sigma(A) = \lambda_1, \lambda_2, ..., \lambda_1$ and their algebric multiplicities $m_a(\lambda_1), m_a(\lambda_2)$, etc.

2. For every $\lambda \in \sigma(A)$, we need the eigenspace and the geometric multiplicity of λ using the equation $(A - \lambda I_n)v = 0 \Rightarrow$ eigenspace $m_g(\lambda)$ is the dimension of the eigenspace

3. For every $\lambda \in \sigma(A)$

   a. If $m_a(\lambda) = m_g(\lambda) = 1 \Rightarrow$ take v eigenvector of λ and create $e^{Ax}v$ solution of SDE

   b. If $m_a(\lambda) = m_g(\lambda) = k \geq 2 \Rightarrow$ take $v_1, v_2, ..., v_n$ linearly independent vectors

   c. If $m_a(\lambda) > m_g(\lambda)$, but $m_g(\lambda) = 1 \Rightarrow$ take v eigenvectors of λ and create principle vectors

      ex. $(A - \lambda I_n)v_{p_1} = v$

      $(A - \lambda i_n)v_{p_2} = v_{p_1}$

4. Write the general solution

Example:

Determine the general solution of the following SDE, $Y' = AY$:

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{pmatrix}$$

$$P(\lambda) = \det(A - \lambda I_3) = 0 \Rightarrow \begin{vmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 3 & 3 & 4-\lambda \end{vmatrix} = 0$$

$$= (7-\lambda) \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 3 & 3 & 4-\lambda \end{vmatrix} = (7-\lambda) \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1-\lambda & 0 \\ 3 & 0 & 1-\lambda \end{vmatrix}$$

$$= (7-\lambda)(1-\lambda)^2 \Rightarrow \lambda_1 = 7 \Rightarrow ma(A) = 1$$
$$\lambda_2 = 1 \Rightarrow ma(A) = 2$$

$$V(A) = \{7, 1\} \in \mathbb{R}$$

$\lambda_1 = 7 \Rightarrow (A - \lambda_1 I_3) \cdot V = 0 \Rightarrow \begin{pmatrix} -5 & 1 & 1 \\ 2 & -4 & 2 \\ 3 & 3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$\Rightarrow \begin{cases} -5x + y + z = 0 \Rightarrow -4x + 2y = 0 \Rightarrow y = 2x \\ 2x - 4y + 2z = 0 \\ 3x + 3y - 3z = 0 \Rightarrow z = x + y = 3x \end{cases}$

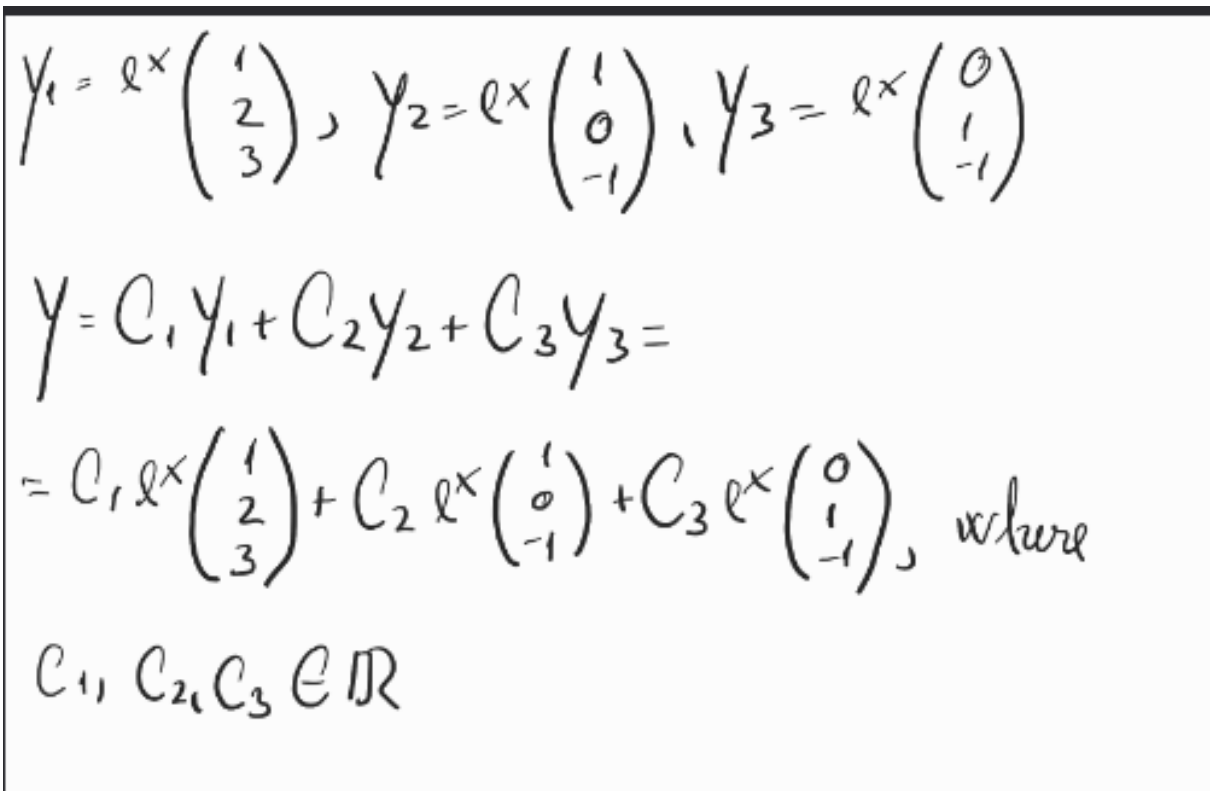
$V_1 = \begin{pmatrix} x \\ 2x \\ 3x \end{pmatrix} = x \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \; mg(\lambda_1) = 1 \Rightarrow mg = ma$

$\lambda_2 = 1 \Rightarrow (A - \lambda_2 I_3) V = 0 \Rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$\Rightarrow \begin{cases} x + y + z = 0 \Rightarrow z = -x - y \\ 2x + 2y + 2z = 0 \\ 3x + 3y + 3z = 0 \end{cases}$

$V_2 = \begin{pmatrix} x \\ y \\ -x - y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \; mg(\lambda_2) = 2 \Rightarrow$

$\Rightarrow ma = mg$



Systems of differential equations, with constant coefficients, nonhomogeneous

Algorithm:

1. Determine the eigenvalues of the matrix A using the formula $det(A - \lambda I_n) = 0$
2. Find e^{Ax}
3. We have two cases:
   a. Case I($\lambda_1 \neq \lambda_2$):
      i. Find v_1 eigenvector for λ_1 and v_2 eigenvector for λ_2
      ii. Form the matrix $C = (v_1, v_2)$ and find the matrix C^{-1}
      iii. We will define $e^{Dx} = \begin{bmatrix} e^{\lambda_1 x} & 0 \\ 0 & e^{\lambda_2 x} \end{bmatrix}$ and $e^{Ax} = Ce^{Dx}C^{-1}$
   b. Case II($\lambda, ma = 2$):
      i. Find v eigenvector of λ and create v_{p1}
      ii. Form $C = (v, v_{p1})$ and determine C^{-1}

iii. We will define $e^{Jx} = e^{\lambda x}\begin{bmatrix} x & 1 \\ 0 & 1 \end{bmatrix}$ and $e^{Ax} = Ce^{Jx}C^{-1}$

4. Calculate $Y_h = e^{Ax}Y_0$
5. Find $Y_p = \int_0^x e^{A(x-t)}b(t)dt$
6. Finally, write the solution $Y = Y_h + Y_p$

Example:

$$\begin{cases} y_1' = -2y_1 + 2y_2 \\ y_2' = y_1 - y_2 + e^{-3x} \end{cases}, \; y_1(0) = 2, y_2(0) = 1;$$

a) We have that $A = \begin{bmatrix} -2 & 2 \\ 1 & -1 \end{bmatrix}$, $B(x) = \begin{bmatrix} 0 \\ e^{-3x} \end{bmatrix}$, $x_0 = 0$ and $Y_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

[*Stage 1*]: We need to find e^{Ax}. The eigenvalues of the matrix A are $\lambda_1 = 0$ and $\lambda_2 = -3$, hence we have a diagonalizable matrix (see E 3. a)).

We get that

$$e^{Dx} = \begin{bmatrix} 1 & 0 \\ 0 & e^{-3x} \end{bmatrix}$$

and

$$e^{Ax} = \frac{1}{3}\begin{bmatrix} 1 + 2e^{-3x} & 2 - 2e^{-3x} \\ 1 - e^{-3x} & 2 + e^{-3x} \end{bmatrix}.$$

[*Stage 2*]: Now we compute $Y_h = e^{Ax}Y_0$, the solution of the associated homogenous system. We obtain that

$$Y_h = \frac{1}{3}\begin{bmatrix} 1 + 2e^{-3x} & 2 - 2e^{-3x} \\ 1 - e^{-3x} & 2 + e^{-3x} \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 4 + 2e^{-3x} \\ 4 - e^{-3x} \end{bmatrix}.$$

[*Stage 3*]: We proceed to determining a particular solution of the initial system: $Y_p = \int_0^x e^{A(x-t)}B(t)\mathrm{d}t$. We have that

$$\begin{aligned} Y_p &= \int_0^x \frac{1}{3}\begin{bmatrix} 1+2e^{-3(x-t)} & 2-2e^{-3(x-t)} \\ 1-e^{-3(x-t)} & 2+e^{-3(x-t)} \end{bmatrix}\begin{bmatrix} 0 \\ e^{-3t} \end{bmatrix}\mathrm{d}t \\ &= \frac{1}{3}\int_0^x \begin{bmatrix} 2e^{-3t}-2e^{-3x} \\ 2e^{-3t}+e^{-3x} \end{bmatrix}\mathrm{d}t \\ &= \frac{1}{3}\begin{bmatrix} \dfrac{2e^{-3t}}{-3}\Big|_0^x - 2e^{-3x}t\Big|_0^x \\ \dfrac{2e^{-3t}}{-3}\Big|_0^x + e^{-3x}t\Big|_0^x \end{bmatrix} \\ &= \frac{1}{3}\begin{bmatrix} \frac{2}{3}-\frac{2}{3}e^{-3x}-2xe^{-3x} \\ \frac{2}{3}-\frac{2}{3}e^{-3x}+xe^{-3x} \end{bmatrix}. \end{aligned}$$

[*Stage 4*]: Now we add up the results from the previous two stages: $Y = Y_h + Y_p$. In conclusion,

$$Y = \frac{1}{3}\begin{bmatrix} \frac{14}{3}+\frac{4}{3}e^{-3x}-2xe^{-3x} \\ \frac{14}{3}-\frac{5}{3}e^{-3x}+xe^{-3x} \end{bmatrix}.$$

Stability

For the stability exercises we have only to find if the system is **asymptotically stable**, but for theory we have to learn all the cases, so be carefull!

Remember, for this type of exercises we have **3 methods** and for the partial exam we have to use all of them, so learn all of the methods.

Method 1:

1. Find the eigenvalues of the matrix A, using the formula $det(A - \lambda I_n) = 0$
2. If for all the eigenvalues λ we have $Re(\lambda) < 0$, then the system is asymptotically stable

Method 2(Routh-Hurwitz Criterion):

1. Calculate $P_A(\lambda) = det(\lambda I_n - A) = a_0\lambda^n + a_1\lambda^{n-1} + ... + a_n$
2. Check that $a_j > 0$ to create the Hurwitz matrix $H_n = \begin{bmatrix} a_1 & a_0 & 0 & ... & 0 & & \\ a_3 & a_2 & a_1 & a_0 & 0 & ... & 0 \\ ... & ..... & ..... & ..... & ..... & & \\ a_{2n-1} & a_{2n-2} & ... & a_0 & & & \end{bmatrix}$
3. If the H_n(Hurwitz matrix) is a positive defined matrix, then the system is asymptotically stable

Remark: we can remember from Linear Algebra that a matrix is positive defined by using Sylvester Criterion, where all the minors **must be positive!**

Method 3(Lyapunov equation):

If, for a certain matrix Q, **symmetric and positive defined**, we can find a matrix P, also symmetric and positive defined, so that we can use the so called Lyapunov equation as follows: $A^TP + PA = -Q$

If this equation is verified, we can deduce that the system is asymptotically stable.

Remark: the matrix Q will be always given in problems.

Example:

Check the stability of the system $Y' = AY$ using all the three methods:

$A = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix}$, $Q = \begin{pmatrix} 16 & 0 \\ 0 & 2 \end{pmatrix}$

Method 1:

$P(\lambda) = \det(A - \lambda I_2) = 0 \Rightarrow \begin{vmatrix} -2-\lambda & 1 \\ 0 & -2-\lambda \end{vmatrix} = 0$

$\Rightarrow (-2-\lambda)^2 = 0 \Rightarrow \lambda = -2$

$Re(\lambda) = -2 < 0 \Rightarrow$ the system is asymptotically stable

Method 2:

$P(\lambda) = \det(\lambda I_2 - A) = \lambda^2 + 4\lambda + 4$

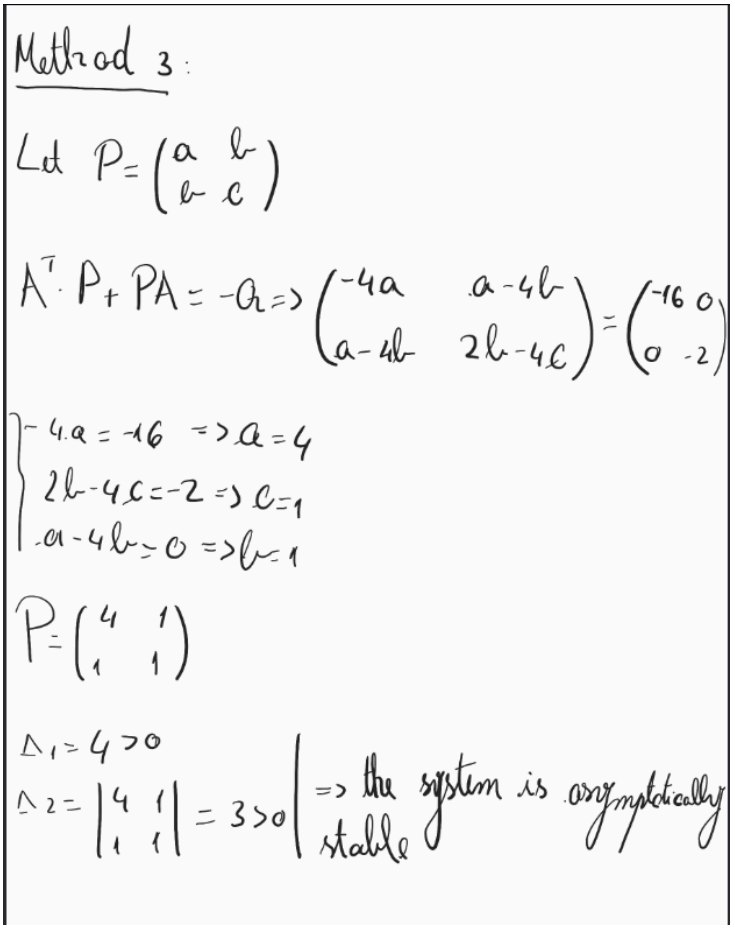
$a_2 = 1, a_1 = 4, a_0 = 4$

We will build the Hurwitz matrix $\Rightarrow$

$\Rightarrow H_2 = \begin{pmatrix} a_1 & a_0 \\ a_3 & a_2 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 0 & 1 \end{pmatrix}$

$\Delta_1 = 4 > 0$

$\Delta_2 = \begin{vmatrix} 4 & 4 \\ 0 & 1 \end{vmatrix} = 4 > 0 \Rightarrow$ the matrix H_2 is positive defined $\Rightarrow$ the system is asymptotically stable



Finding the derivability in one point(Cauchy-Riemann conditions)

Algorithm:

1. Replace $z = x + iy$ and find $Re(f) = u(x, y)$ and $Im(f) = v(x, y)$
2. Compute $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$, $\frac{\partial u}{\partial y}$
3. Check if the Cauchy-Riemann conditions are true:
   a. $\frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0)$
   b. $\frac{\partial u}{\partial y}(x_0, y_0) = -\frac{\partial v}{\partial x}(x_0, y_0)$

Restoration of an analytic function

I. We know $Re(f) = u(x, y)$

1. Verify if $u(x,y)$ is a harmonic function by $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$
    a. If is not 0, then we will stop here, because we don`t have a harmonic function
    b. If is 0, we will continue the algorithm
2. Write $f`(z) = \frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}$
3. We will impose $y = 0$ such that $f`(x) = \frac{\partial u}{\partial x}(x,0) - i\frac{\partial u}{\partial y}(x,0) = A(x)$
4. Make $f(x) = \int A(x)dx + C$
5. Use the initial condition, where we replace x with z and find the constant C

II. We know $Im(f) = v(x,y)$

Remark: all steps are the same, the only difference is that $f`(z) = \frac{\partial v}{\partial y} + i\frac{\partial v}{\partial x}$

Example:

$$\text{Im } f = y + \arctan\left(\frac{y}{x}\right),\ (x,y) \neq (0,0),\ f(i) = i.$$

b) Let us prove that $v(x,y) = \text{Im } f = y + \arctan\left(\frac{y}{x}\right)$ is a harmonic function. For that, we calculate $\Delta v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}$. Since

$$\frac{\partial v}{\partial x} = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \left(\frac{y}{x}\right)'_x = \frac{-y}{x^2 + y^2},$$

$$\frac{\partial v}{\partial y} = 1 + \frac{1}{1+\left(\frac{y}{x}\right)^2} \cdot \left(\frac{y}{x}\right)'_y = 1 + \frac{x}{x^2+y^2},$$

$$\frac{\partial^2 v}{\partial x^2} = \frac{y}{(x^2+y^2)^2} \cdot 2x = \frac{2xy}{(x^2+y^2)^2},$$

$$\frac{\partial^2 v}{\partial y^2} = \frac{-x}{(x^2+y^2)^2} \cdot 2y = \frac{-2xy}{(x^2+y^2)^2},$$

we get that $\Delta v = 0$, hence the necessary condition that v is harmonic is fulfilled.

Step 1. Calculate $f'(z) = \dfrac{\partial v}{\partial y}(x,y) + i\dfrac{\partial v}{\partial x}(x,y)$ ((3.5) iii)). We have that

$$f'(z) = 1 + \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2};$$

Step 2. Replace y with 0 in $f'(z)$. Then $z = x + iy$ becomes $z = x$ and

$$f'(x) = 1 + \frac{1}{x};$$

Step 3. Integrate $f'(x)$ with respect to x. It follows that

$$f(x) = \int \left(1 + \frac{1}{x}\right) \mathrm{d}x = x + \ln x + C,$$

where C is a constant;

Step 4. Replace x by z, thus $f(z) = z + \operatorname{Ln} z + C$;

Step 5. Replace z with i, so $C = -\operatorname{Ln} i$.

The analytic function is $f(z) = z + \operatorname{Ln} z - \operatorname{Ln} i$, $z \neq 0$.